AIC02
Voltage References and Bias Circuits

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Rebuild the divider, threshold self-bias and startup-assisted references from *AIC Lab.pdf*, PDF pages 22–30, and examine supply and temperature sensitivity. The model library is supplied separately by the user. The 1.0 V supply is a teaching adaptation, not a certification of the model voltage rating. Numerical plots for the handout's original 3.3 V process are not used directly as pass thresholds. The circuits are rebuilt from the handout topology; the earlier code is used only to review dimensions and omissions.

2 Specification Mapping

S denotes actual simulation under stated adapted conditions; D denotes analytical discussion; L denotes a numerical or validation limitation. D01 etc. identify filename prefixes.

Requirement and PDF pages	Deck / check	Status
Two-MOS divider and analytical comparison (22–23)	D01; DC, line AC	S+D
Three-MOS dual references, sizing, isolated well (24)	D02; three sizes	S+D
Threshold self-bias, branch conditions, zero-current solution (25)	D03; two R values	S+D
Startup function and injection (26–27)	D04; ramp and disconnect	S+D
Current setting, self-bias and complete Vbias (27–29)	D05–06; DC/AC/transient	S+D
Process compensation and temperature design (30)	D05–06; -20/27/85 C	S+D/L
Old 0.8/2.1 V and about 1.62 V examples (24,29)	1 V adaptation	L

3 Circuit and Design Derivation

$$\beta_n(V_{ref}-V_{Tn})^2=\beta_p(V_{DD}-V_{ref}-|V_{Tp}|)^2,\quad V_{ref}\simeq\frac{V_{DD}-|V_{Tp}|+\sqrt{\beta_n/\beta_p}V_{Tn}}{1+\sqrt{\beta_n/\beta_p}}.$$

$$I_{threshold} = \frac{V_{GS1}}{R}, \qquad I_{bias} \simeq \frac{2}{\beta R_b^2}\left(1 - \frac{1}{\sqrt{K}}\right)^2, \qquad V_{bias} = \left(1 + \frac{R_3}{R_4}\right) V_{GSn} - |V_{SGp}|.$$

These are strong-inversion, long-channel approximations for matched branches. β must belong to the device polarity in the actual resistor-degenerated branch. The μ_n printed on PDF page 27 cannot be inserted directly into the PMOS branch of page 28; K here is the corresponding PMOS geometry ratio. Body effect, finite output conductance and leakage remain in the supplied model. Old-process constants are not transplanted into the numerical results.

D01 retains $W_n/W_p/L = 21/10/0.3\,\mu$m and corrects the grounded PMOS bulk. Separate cases use $W_p = 20\,\mu$m or $W_n = 42\,\mu$m. D02 retains two $20/0.3\,\mu$m PMOS devices and a $17.5/0.3\,\mu$m NMOS, then doubles either PMOS or NMOS widths. The middle PMOS body-source tie requires an isolated N-well in layout, with its area and isolation rules.

D03 restores the omitted M5 mirror branch, uses $R = 5/10$ kΩ, and holds the output at 0.5 V. M2's NMOS bulk is grounded, retaining body effect; a body-source tie would require confirmed well isolation. `.nodeset` is a solver starting hint, not hardware startup.

D04–06 implement MS1–MS3 and the four-MOS core. The old supply sine starting at 1 V is replaced by a PWL rise from zero. The complete circuit uses $R_b = 1$ kΩ and $R_3/R_4 = 5/1$ kΩ; D05 also tests $R_3 = 3$ kΩ and $R_b = 2$ kΩ.

4 Simulation Setup

Runs use PrimeSim HSPICE S-2021.09 win64, a baseline temperature of 27 °C, `runlvl=5`, `accurate`, and `.op/.print/.measure` outputs. Every `.alter` resets all baseline parameters and temperature to prevent state from carrying into the next case. Actual runs completed without an error or failed measurement. Suppressed model warnings remain, so a warning-free run is not claimed. Public netlists contain no model paths. This comment marks the model setup required locally before execution; the model and its configuration remain local:

```
* MODEL_SETUP_REQUIRED
```

CSV files and numerical summaries are in the lab's `results/` directory. Every figure is redrawn from actual printed data, not a waveform-viewer screenshot. `manifest.json` supplies specification mapping and case definitions.This lab contains six decks and 22 cases. D01–03 sweep supply voltage and measure the 1 Hz supply-to-output gain. D04 uses 1/30 μs ramps and a 50 μs window. D06 observes 100 μs with a 20 ns maximum step and UIC zero-charge initialization. The reported $t_{1\%}$ starts when the supply ramp begins and requires every later sample to remain within 1% of the final value. This is a report-defined criterion, not a handout guarantee.

5 Results and Specification Checks

Case	V_{ref1} (V)	V_{ref2} (V)	dV_o/dV_{DD}	P_b (μW)
D01/C0	0.403	–	0.273	99.87
D01/C1	0.432	–	0.316	163.03
D01/C2	0.375	–	0.239	119.20
D02/C0	0.300	0.650	–	–
D02/C1	0.322	0.661	–	–
D02/C2	0.278	0.639	–	–

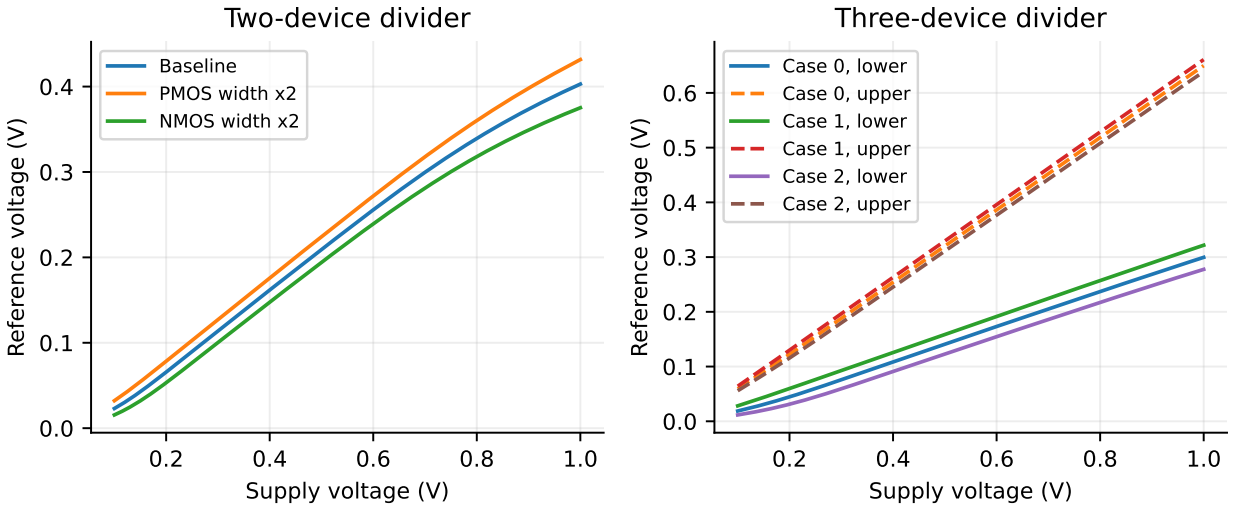


Figure 1: Actual supply sweeps of both divider references. The old 2.1 V upper reference is unattainable with the present 1 V supply.

Case	Temp. (°C)	V_{ref} (V)	I_{DD} (μA)	$\lvert A_{supply}\rvert$
D05/C0	27	0.44410	311.95	0.01283
D05/C1	-20	0.37291	278.86	3.02680
D05/C2	85	0.61066	420.07	0.47272
D05/C3	27	0.21186	313.62	0.05349
D05/C4	27	0.22662	203.08	1.97827

Case	Ramp (μs)	V_{final} (V)	$t_{1\%}$ (μs)	P_b (μW)
D06/C0	1	0.44408	4.18	311.94
D06/C1	30	0.44410	27.82	311.95
D06/C2	1	0.37291	12.72	278.86
D06/C3	1	0.61066	1.80	420.07
D06/C4	1	0.44404	4.18	311.92

Case	$I_{inj,pk}$ (nA)	$I_{inj,final}$ (nA)	I_{core} (μA)
D04/C0	222.739	3.8605	103.102
D04/C1	3.978	3.8009	103.098
D04/C2	0.087	0.0000	103.089
D04/C3	0.023	0.0000	103.090

D01 follows the analytical trend: stronger relative PMOS drive raises the output; stronger NMOS drive lowers it. D03 at 5 kΩ gives $V_{ref} = 0.39866$ V, so $V_{ref}/R = 79.732\,\mu$A; M5 measures approximately $81.000\,\mu$A. Geometry mismatch, unequal V_{DS} and body effect must be considered in that difference.

D05/C0 is baseline; C1/C2 are -20/85 °C; C3 uses $R_3 = 3$ kΩ; C4 uses $R_b = 2$ kΩ. D06/C0–C4 are baseline, slow ramp, cold, hot and disconnected startup. In D04/C2–C3 the core still starts in these nominal simulations with the injection path disconnected. This observation is retained and does not show that startup is unnecessary for every ramp and corner.

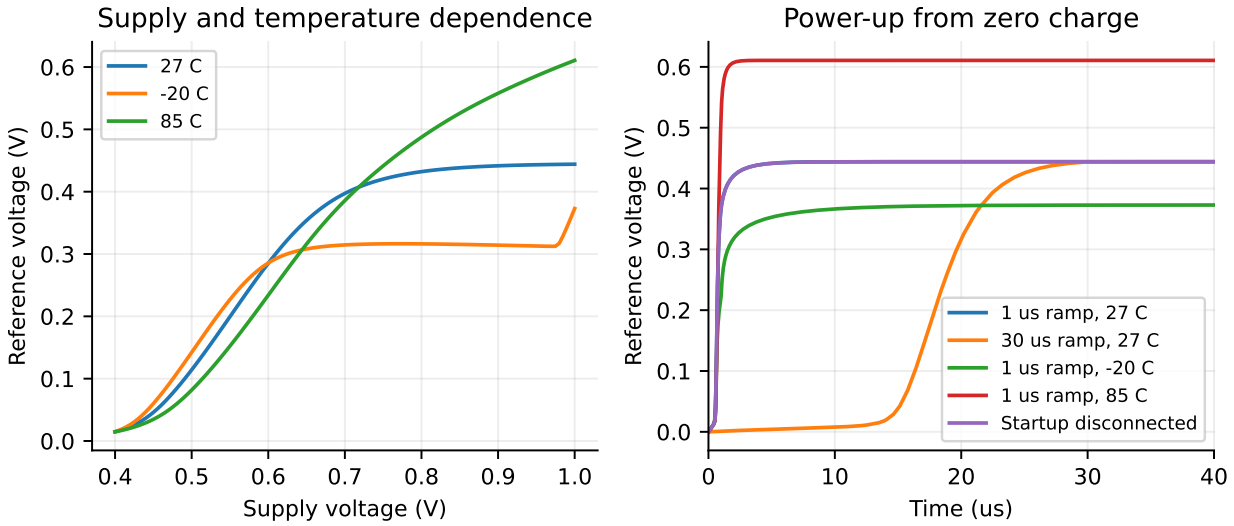


Figure 2: Complete-reference DC temperature sweeps and startup. Temperature drift and the cold DC branch transition are retained.

6 Discussion

Zero current is also a self-bias equilibrium. Near $I = 0$, division and the strong-inversion derivation of the nonzero solution are not valid. Startup injects temporary current to leave that state and should then weaken to avoid disturbing the reference. The fast ramp gives approximately 222.7 nA peak injection and 3.86 nA at steady state. The slow ramp lacks the same spike, demonstrating ramp and trajectory dependence. Nominal startup without the injection path can result from leakage and numerical initial perturbations; it is not a cross-corner startup design.

PDF page 29 says approximately 1.62 mV while the plot is near 1.62 V. This is treated as a unit ambiguity, and neither value is imposed as a required 1 V result. Divider references depend on threshold, mobility and geometry. Resistor degeneration reduces supply dependence but retains sensitivity to β, resistance and temperature. The complete reference still drifts substantially from -20 to 85 °C and is not fully process- or temperature-compensated. Temperature stabilization would require complementary temperature-coefficient terms, resistor selection and trimming, followed by validation with available corner and temperature models.

7 Conclusion

Every handout topology, startup question and compensation discussion has corresponding code and actual results. A reference near 0.444 V is obtained under the stated 1 V conditions. Original-process amplitude, complete process compensation and cross-corner startup reliability are not claimed as passed.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. All ten MOS devices of `05_complete_reference.sp` are shown in three connected panels. All device lengths are 0.3 μm. Widths below are in μm.

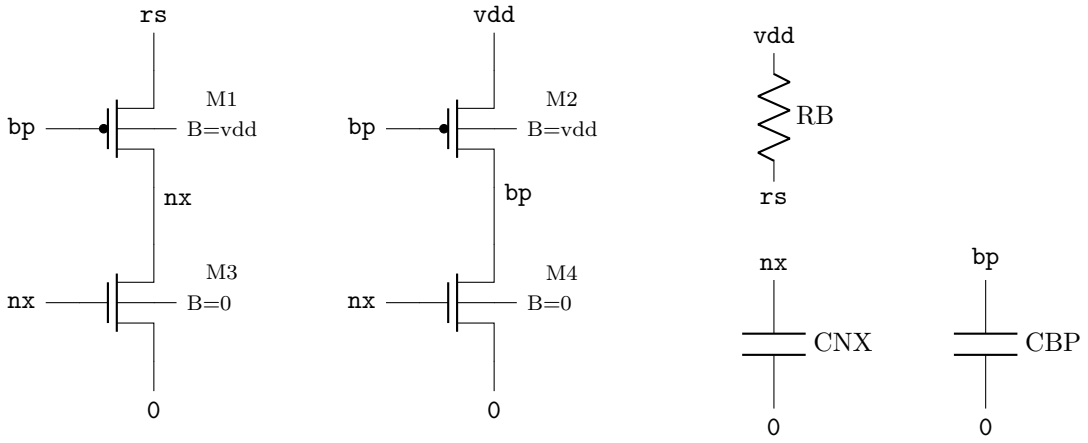


Figure 3: Self-biased reference core: M1=115, M2=22, M3=M4=4.85; RB=rbias. CNX=CBP=20 fF.

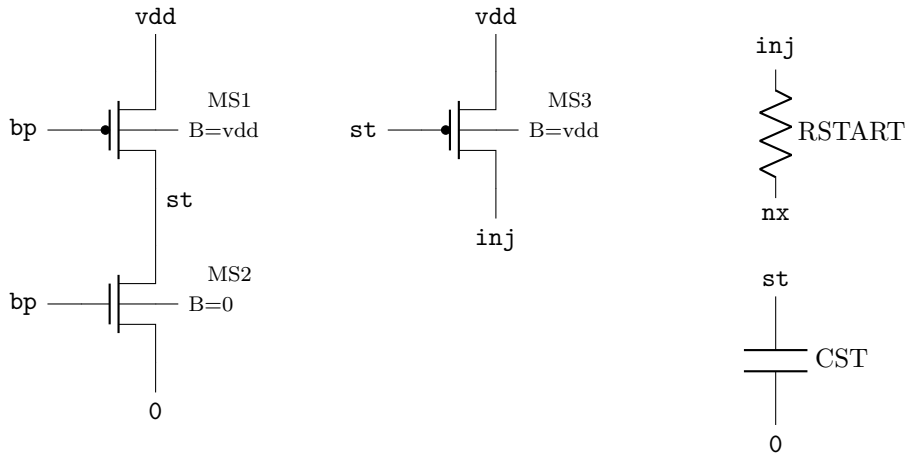


Figure 4: Startup network: MS1=2, MS2=0.2, MS3=10; CST=20 fF, RSTART=rstart.

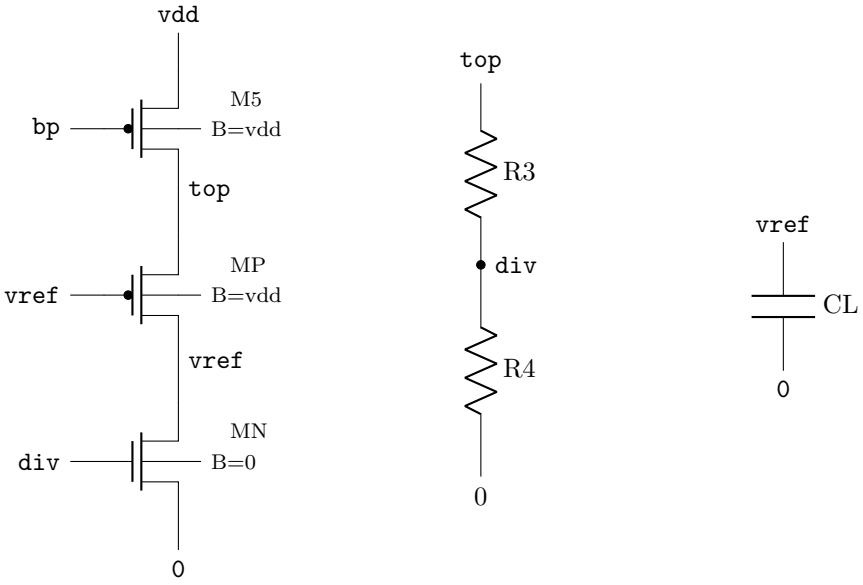


Figure 5: Reference output: M5=MP=22, MN=4.85; R3=rupper, R4=1 kΩ, CL=1 pF.

M1's source is rs and MP's source is top; their bodies remain at vdd. These are separate body/source terminals, not body-to-source shorts. Baseline rbias=1 kΩ, rstart=1 mΩ, rupper=5 kΩ, vdd=1 V. The MOS-only divider and startup experiments use their own source decks; these panels show the combined reference.